Lecture 8

Canonical maps and generating functions

Objectives. *Hamilton's equations survive a far larger freedom of coordinates than Lagrange's: q and p may be mixed. This lecture asks how a map that keeps Hamilton's equations is recognised, and how one is built. It is recognised by its brackets, and equivalently by one explicit 1-form being closed. On a star-shaped set a closed 1-form is the differential of a function, so a single function of $2n$ variables, a* generating function*, determines a canonical map where the required coordinates are independent. A generating function may also depend on time, and then the Hamiltonian changes by $\partial F_2/\partial t$. The lecture ends with the question the next one answers: which F_2 makes the new Hamiltonian vanish?*

By the end of this lecture you will be able to:

- Test whether a coordinate transformation is canonical.
- Distinguish closed forms from exact forms.
- Construct canonical maps from generating functions where the required coordinates are independent.
- Compute the transformed Hamiltonian for a time-dependent generating function.

8.1 Canonical maps

Definition 8.1 (Canonical map)**.** Let $U, V \subseteq \mathbb{R}^{2n}$ be open. A C^∞ map $F\colon U \to V$ is *canonical* if $DF(x) \in \mathrm{Sp}(2n)$ for every $x \in U$.

Theorem 8.2 (Canonical maps, four ways)**.** *For a C^∞ map $F\colon U \to V$ the following are equivalent.*

(i) *F is canonical: $DF(x)^{\mathsf{T}} J\, DF(x) = J$ for all x.*

(ii) *F preserves ω: $\omega\big(DF(x)u, DF(x)v\big) = \omega(u,v)$ for all x, u, v.*

(iii) *F preserves all brackets: $\{f \circ F, g \circ F\} = \{f,g\} \circ F$ for all $f, g \in C^\infty(V)$.*

(iv) *F preserves the brackets of the coordinates: $\{F_a, F_b\} = J_{ab}$ for $a, b = 1, \dots, 2n$; that is, writing $F = (Q,P)$, $\{Q_i, Q_j\} = \{P_i, P_j\} = 0$ and $\{Q_i, P_j\} = \delta_{ij}$.*

Proof. Write $A = DF(x)$. The matrix condition and preservation of ω are equivalent, since $\omega(Au, Av) = u^{\mathsf{T}} A^{\mathsf{T}} J A v$ for all u, v. The chain rule gives $\nabla(f \circ F) = A^{\mathsf{T}} (\nabla f \circ F)$, so by (5.1) in the form $\{f,g\} = \nabla f^{\mathsf{T}} J \nabla g$,

$$\{f \circ F, g \circ F\}(x) = (\nabla f \circ F)^{\mathsf{T}}\, A J A^{\mathsf{T}}\, (\nabla g \circ F)(x). \tag{8.1}$$

Taking f, g to be the coordinate functions of V shows that the matrix with entries $\{F_a, F_b\}$ is AJA^{T}, so the coordinate-bracket condition says $AJA^{\mathsf{T}} = J$. Then (8.1) gives preservation of all brackets, because $\{f, g\} \circ F$ is the same expression with J in the middle. Conversely, preservation of all brackets contains the coordinate-bracket condition. It remains to see that $AJA^{\mathsf{T}} = J$ and $A^{\mathsf{T}}JA = J$ say the same thing. Either forces $(\det A)^2 = 1$, so A is invertible. Inverting $AJA^{\mathsf{T}} = J$ and using $J^{-1} = -J$ gives $A^{-\mathsf{T}}JA^{-1} = J$, that is $A^{-1} \in \mathrm{Sp}(2n)$, and $\mathrm{Sp}(2n)$ is a group (Proposition 6.14), so $A \in \mathrm{Sp}(2n)$. The converse implication is the same argument run backwards. ∎

The coordinate-bracket condition of Theorem 8.2 gives a practical test using $2n(2n-1)/2$ brackets of explicit functions. § 8.3 tests canonicity by one closed 1-form, and § 8.4 builds canonical maps from a single function. By Theorem 6.15, every Hamiltonian flow is canonical.

For example, the map $Q = p$, $P = q$ is not canonical, since $\{Q, P\} = -1$. The map $Q = 2q$, $P = p$ is not canonical either, since $\{Q, P\} = 2$.

Canonical maps keep Hamilton's equations, as we now show.

Proposition 8.3 (Canonical maps keep Hamilton's equations). *Let $F = (Q, P) : U \to V$ be a canonical bijection, let $H \in C^\infty(U)$, and put $K = H \circ F^{-1}$. If $x(t)$ solves Hamilton's equations for H, then $F(x(t))$ solves Hamilton's equations for K in the coordinates (Q, P) of V.*

Proof. The map F has a smooth inverse, because DF is invertible (inverse function theorem). Write Q_i and P_i also for the coordinate functions on V. Then the new position along the motion is $Q_i = Q_i \circ F$, and $H = K \circ F$. By (5.5), preservation of brackets and (5.4),

$$\dot{Q}_i = \{Q_i \circ F, K \circ F\} = \{Q_i, K\} \circ F = \frac{\partial K}{\partial P_i} \circ F.$$

In the same way $\dot{P}_i = \{P_i, K\} \circ F = -(\partial K/\partial Q_i) \circ F$. ∎

8.2 1-forms in coordinates

The test of this lecture is an integral of $p\,\mathrm{d}q$ along curves. That expression takes a *direction* as its argument and returns a number. It is a 1-form.

Definition 8.4 (1-form). Let $U \subseteq \mathbb{R}^N$ be open with coordinates $x = (x_1, \dots, x_N)$. A *1-form* on U is a C^∞ map that assigns to each $x \in U$ a linear functional α_x on $\mathbb{R}^N$. Writing $\mathrm{d}x_i$ for the functional $v \mapsto v_i$, every 1-form is

$$\alpha = \sum_{i=1}^{N} a_i(x)\,\mathrm{d}x_i, \qquad a_i \in C^\infty(U), \qquad \alpha_x(v) = \sum_i a_i(x) v_i. \tag{8.2}$$

The model is the differential of a function: for $f \in C^\infty(U)$, $\mathrm{d}f = \sum_i (\partial f/\partial x_i)\,\mathrm{d}x_i$, so $\mathrm{d}f_x(v)$ is the derivative of f at x in the direction v. Our main example is the *canonical* 1*-form* on a phase space $\mathcal{M} \subseteq \mathbb{R}^{2n}$ with coordinates $x = (q, p)$ (§ 4.1):

$$\theta = p\,\mathrm{d}q = \sum_{i=1}^{n} p_i\,\mathrm{d}q_i. \tag{8.3}$$

It is the boundary term of Proposition 3.10 with the variation replaced by an arbitrary direction.

Definition 8.5 (Line integral). For a piecewise C^1 curve $\gamma\colon [a,b] \to U$,

$$\int_\gamma \alpha := \int_a^b \alpha_{\gamma(t)}\big(\dot\gamma(t)\big)\mathrm{d}t = \int_a^b \sum_i a_i(\gamma(t))\,\dot\gamma_i(t)\,\mathrm{d}t. \tag{8.4}$$

A curve with $\gamma(a) = \gamma(b)$ is a *loop*, and the integral over a loop is written $\oint_\gamma \alpha$.

The chain rule makes (8.4) independent of the parametrisation, as long as its orientation is kept: if $t = \tau(u)$ with $\tau' > 0$ then $\dot\gamma\,\mathrm{d}t$ is unchanged. Reversing the orientation changes the sign. Along a phase path $t \mapsto (q(t), p(t))$,

$$\int_\gamma \theta = \int_a^b p(t)\cdot\dot q(t)\,\mathrm{d}t, \tag{8.5}$$

the first half of the phase-space action (4.10).

Exact, and closed

Definition 8.6 (Exact, closed). The form α is *exact* on U if $\alpha = \mathrm{d}f$ for some $f \in C^\infty(U)$, and *closed* if

$$\frac{\partial a_j}{\partial x_i} = \frac{\partial a_i}{\partial x_j} \qquad \text{for all } i, j. \tag{8.6}$$

Exact forms are the ones whose integrals are path-independent: by the chain rule and the fundamental theorem of calculus,

$$\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a)), \qquad \text{so} \qquad \oint_\gamma \mathrm{d}f = 0 \text{ on every loop.} \tag{8.7}$$

Every exact form is closed, because $a_i = \partial f/\partial x_i$ and mixed partial derivatives of a C^∞ function commute. The converse depends on the domain and is false in general.

Theorem 8.7 (Poincaré lemma for 1-forms). *Let $U \subseteq \mathbb{R}^N$ be open and* star-shaped *about x_0: for every $x \in U$ the segment from x_0 to x lies in U. Then every closed 1-form on U is exact. Explicitly, $\alpha = \mathrm{d}f$ with*

$$f(x) = \int_0^1 \sum_i a_i\big(x_0 + s(x - x_0)\big)\,(x_i - x_{0,i})\,\mathrm{d}s. \tag{8.8}$$

Proof. Take $x_0 = 0$, which costs only a translation. The integrand of (8.8) is C^∞ in (s, x) on $[0,1]\times U$, so f is C^∞ and may be differentiated under the integral sign:

$$\frac{\partial f}{\partial x_j}(x) = \int_0^1 \Big[a_j(sx) + s\sum_i \frac{\partial a_i}{\partial x_j}(sx)\,x_i\Big]\mathrm{d}s.$$

Closedness replaces $\partial a_i/\partial x_j$ by $\partial a_j/\partial x_i$, and then the bracket is exactly

$$a_j(sx) + s\sum_i \frac{\partial a_j}{\partial x_i}(sx)\,x_i = \frac{\mathrm{d}}{\mathrm{d}s}\big[s\,a_j(sx)\big].$$

Integrating from 0 to 1 gives $a_j(x)$. ∎

Closedness entered once, in the middle of the proof. The hypothesis on the domain cannot simply be omitted. On the punctured plane $U = \mathbb{R}^2 \setminus \{0\}$, which is not star-shaped, put

$$\alpha = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}. \tag{8.9}$$

The two cross derivatives agree:

$$\frac{\partial}{\partial x}\frac{x}{x^2+y^2} = \frac{y^2 - x^2}{(x^2+y^2)^2} = \frac{\partial}{\partial y}\frac{-y}{x^2+y^2}.$$

Thus the form is closed. It is not exact. On the unit circle $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$, the integrand is $\sin^2 t + \cos^2 t = 1$ and $\oint_\gamma \alpha = 2\pi \neq 0$, which (8.7) forbids for an exact form. On any open half-plane whose boundary passes through the origin, for instance $x > 0$, it *is* exact: there $\alpha = \mathrm{d}\arctan(y/x)$. The form is the differential of the polar angle, which exists only locally, and it is written $\mathrm{d}\vartheta$ although no such function ϑ exists on U. We use ϑ for the angle because θ denotes the canonical 1-form.

Substituting a map

If $F\colon V \to U$ is C^∞, a 1-form α on U can be rewritten in the coordinates y of V: substitute $x = F(y)$ and $\mathrm{d}x_i = \sum_b (\partial F_i/\partial y_b)\,\mathrm{d}y_b$. The result is the 1-form

$$F^*\alpha = \sum_b \Big(\sum_i a_i(F(y)) \frac{\partial F_i}{\partial y_b} \Big) \mathrm{d}y_b \qquad \text{on } V, \tag{8.10}$$

and by the chain rule, for every curve γ in V,

$$\int_{F\circ\gamma} \alpha = \int_\gamma F^*\alpha. \tag{8.11}$$

Thus (8.10) is the change-of-variables rule for line integrals.

8.3 The closedness test

The four tests of Theorem 8.2 check a map; they do not build one. This section adds a fifth test, which does. For $n = 1$ canonical means $\{Q, P\} = 1$, that is $\det DF \equiv 1$: $F = (Q, P)$ preserves oriented area. Area is what Green's theorem turns into a loop integral of $p\,\mathrm{d}q$: for a loop γ bounding a region D,

$$\oint_\gamma p\,\mathrm{d}q = -\operatorname{area}(D) \tag{8.12}$$

when γ runs anticlockwise, and $+\operatorname{area}(D)$ when it runs clockwise — which is the sense in which the oscillator's own orbits are traversed (Figure 6.1). For the orbit $H = \frac{p^2}{2m} + \frac{1}{2}m\nu^2 q^2 = E$, an ellipse with semi-axes $\sqrt{2E/m\nu^2}$ and $\sqrt{2mE}$,

$$\oint p\,\mathrm{d}q = \pi\sqrt{\frac{2E}{m\nu^2}}\sqrt{2mE} = \frac{2\pi E}{\nu}, \tag{8.13}$$

a number that Lecture 9 will call $2\pi I$ (Figure 8.1). So area preservation should show in the loop integrals of $F^*\theta - \theta$, which compare $\oint_{F\circ\gamma} p\,\mathrm{d}q$ with $\oint_\gamma p\,\mathrm{d}q$ (8.11). The theorem makes this a closedness condition, in any number of degrees of freedom.

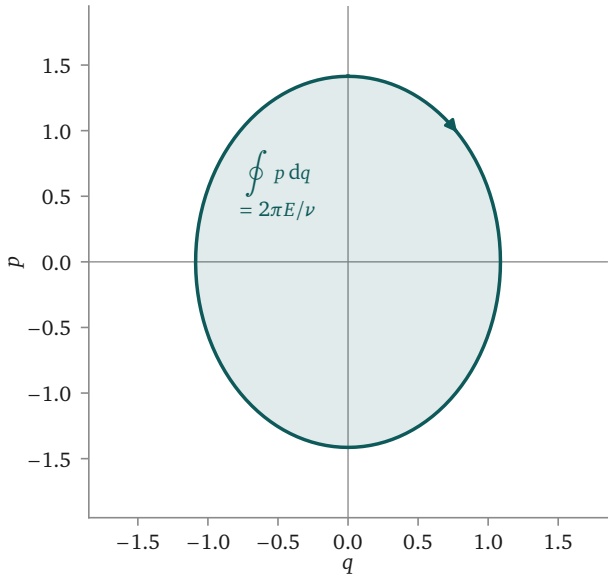


Figure 8.1: One orbit of the oscillator, $m = 1$, $\nu = 1.3$, $E = 1$, traversed clockwise, the sense of its flow. In that sense $\oint p\,\mathrm{d}q$ is + the shaded area, $2\pi E/\nu$ (8.13).

Theorem 8.8 (Canonical means one 1-form is closed)**.** *Let $U \subseteq \mathbb{R}^{2n}$ be open and $F = (Q, P) : U \to \mathbb{R}^{2n}$ be C^∞. Write*

$$\lambda_F := F^*\theta - \theta = \sum_{i=1}^{n} P_i\,\mathrm{d}Q_i - \sum_{i=1}^{n} p_i\,\mathrm{d}q_i, \tag{8.14}$$

a 1-form on U, in which $P_i\,\mathrm{d}Q_i$ means $P_i(q,p)\sum_a (\partial Q_i/\partial x_a)\,\mathrm{d}x_a$. Then the closedness defect of λ_F is

$$\frac{\partial(\lambda_F)_b}{\partial x_a} - \frac{\partial(\lambda_F)_a}{\partial x_b} = -\big(DF^{\mathsf{T}} J\,DF - J\big)_{ab}, \tag{8.15}$$

so F is canonical if and only if λ_F is closed*, that is, if and only if $p\cdot\mathrm{d}q - P\cdot\mathrm{d}Q$ is closed.*

Proof. Only (8.15) needs proof. For the first sum in (8.14), whose ath coefficient is $\sum_i P_i\,\partial_a Q_i$, the second derivatives cancel in the antisymmetric combination:

$$\partial_a\Big(\sum_i P_i\partial_b Q_i\Big) - \partial_b\Big(\sum_i P_i\partial_a Q_i\Big) = \sum_i\big(\partial_a P_i\,\partial_b Q_i - \partial_a Q_i\,\partial_b P_i\big) = -\big(DF^{\mathsf{T}} J\,DF\big)_{ab},$$

by the definition of J in the coordinate order $(q_1,\dots,q_n,p_1,\dots,p_n)$. Applying this with F the identity gives the corresponding statement for θ itself, namely $-J_{ab}$, and subtracting gives (8.15). Closedness of λ_F is the vanishing of the left side for all a, b, and $DF^{\mathsf{T}}JDF = J$ is Definition 8.1. ∎

For $n = 1$ the single independent entry of (8.15) is $1 - \{Q, P\}$: the criterion agrees with the coordinate-bracket test of Theorem 8.2. For the rotation (6.2), $\{Q, P\} = \cos^2 s + \sin^2 s = 1$, so its λ_F is closed; Example 8.11 gives its generating functions. Negating a 1-form does not change whether it is closed, so the criterion may be read off either of λ_F and $-\lambda_F = p\cdot\mathrm{d}q - P\cdot\mathrm{d}Q$.

8.4 Generating functions

Theorem 8.8 turns the construction of canonical maps into the construction of closed 1-forms, and on a star-shaped set a closed 1-form is the differential of a function. One function of $2n$ variables can therefore determine a canonical map. We must specify which variables are independent and where the resulting map is defined.

Theorem 8.9 (Type-1 generating functions)**.**

1. *Let $U \subseteq \mathbb{R}^{2n}$ be open and star-shaped, let $F = (Q, P)$ be canonical on U, and suppose that $\Psi(q,p) = \big(q, Q(q,p)\big)$ is a diffeomorphism of U onto an open $W \subseteq \mathbb{R}^{2n}$ — equivalently, that Ψ is injective and $\det(\partial Q/\partial p) \neq 0$ on U. Then there is $F_1 \in C^\infty(W)$ with*
$$p = \frac{\partial F_1}{\partial q}(q,Q), \qquad P = -\frac{\partial F_1}{\partial Q}(q,Q) \tag{8.16}$$
at corresponding points, and F_1 is unique up to an additive constant.
2. *Conversely, let $F_1 \in C^\infty(W)$ with $\det\big(\partial^2 F_1/\partial q\,\partial Q\big) \neq 0$ at a point $(q_0, Q_0) \in W$, and put $p_0 = \partial F_1/\partial q(q_0, Q_0)$. Then on a neighbourhood of (q_0, p_0) the relations (8.16) determine $Q = Q(q,p)$ and $P = P(q,p)$, and the map $F = (Q,P)$ so defined is canonical.*

Proof. (1) By Theorem 8.8 the 1-form $-\lambda_F = p\cdot \mathrm{d}q - P\cdot \mathrm{d}Q$ is closed on U, hence exact (Theorem 8.7): it is $\mathrm{d}S$ for some $S \in C^\infty(U)$, determined up to a constant. Put $F_1 = S \circ \Psi^{-1} \in C^\infty(W)$. The form $p\cdot\mathrm{d}q - P\cdot\mathrm{d}Q$ is already written in the differentials of the coordinates (q,Q) of W, so its components in those coordinates are p and $-P$; equating them with those of $\mathrm{d}F_1$ gives (8.16).

(2) The map $(q,Q) \mapsto \big(q, \partial F_1/\partial q(q,Q)\big)$ has Jacobian in block form $\left(\begin{smallmatrix} I & 0 \\ * & \partial^2 F_1/\partial q\partial Q \end{smallmatrix}\right)$, whose determinant is $\det(\partial^2 F_1/\partial q\,\partial Q) \neq 0$; by the inverse function theorem it is a diffeomorphism near the point, which is what solves the first group of (8.16) for $Q = Q(q,p)$. The second group then defines P, and by construction $p\cdot\mathrm{d}q - P\cdot\mathrm{d}Q = \mathrm{d}F_1$ there. An exact form is closed, so F is canonical by Theorem 8.8. ∎

In this course *canonical* always means $DF^{\mathsf{T}} J DF = J$, with multiplier 1; older physics usage (Goldstein) also admits $DF^{\mathsf{T}} J DF = cJ$ for a constant $c \neq 0$, which nothing here uses.

The identity map shows at once that (8.16) cannot be the whole story: for it $Q = q$, so (q,Q) are not independent and $\partial Q/\partial p = 0$. The remedy is a Legendre transform in the pair (Q,P) — exactly the trade of Lecture 4, now performed on the new variables: $F_1 = F_2 - Q\cdot P$, where F_2 is a function of (q,P).

Proposition 8.10 (Type-2 generating functions)**.** *Let F_2 be C^∞ on an open subset of the (q,P) space with $\det\big(\partial^2 F_2/\partial q\,\partial P\big) \neq 0$ at a point. Then near that point the relations*
$$p = \frac{\partial F_2}{\partial q}(q,P), \qquad Q = \frac{\partial F_2}{\partial P}(q,P) \tag{8.17}$$
determine a canonical map $(q,p) \mapsto (Q,P)$.

Proof. The map $(q,P) \mapsto \big(q, \partial F_2/\partial q(q,P)\big)$ has Jacobian determinant $\det(\partial^2 F_2/\partial q\,\partial P) \neq 0$, so as in the converse construction of Theorem 8.9, it inverts near the point and gives $P = P(q,p)$. The coordinate Q then comes from the second relation. There (8.17) says $\mathrm{d}F_2 = p\cdot\mathrm{d}q + Q\cdot\mathrm{d}P$, hence
$$p\cdot\mathrm{d}q - P\cdot\mathrm{d}Q = \mathrm{d}\big(F_2 - Q\cdot P\big),$$

which is exact, hence closed, so F is canonical by Theorem 8.8. ∎

The standard list

Throughout, h denotes a diffeomorphism of one open subset of Q onto another, as in Theorem 3.14 and (6.7).

Example 8.11 (Four generating functions).

1. *The identity:* $F_2 = q \cdot P$ gives $p = P$ and $Q = q$. No F_1 generates it.
2. *Point transformations:* $F_2 = h(q) \cdot P$ gives $Q = h(q)$ and $p = Dh(q)^{\mathsf{T}} P$, that is $P = Dh(q)^{-\mathsf{T}} p$. This is precisely the covector law (4.2) and the change of patch (4.3); taking a one-parameter group h^s recovers the cotangent lift (6.7). So every change of coordinates on Q, lifted, is canonical: these are the maps that Theorem 3.6 already allowed on the Lagrangian side.
3. *The swap:* $F_1 = q \cdot Q$ gives $Q = p$ and $P = -q$. Position and momentum exchange roles, with a sign; no F_2 generates it, since $P = -q$ leaves (q, P) dependent. Lecture 4 listed this map among the gains of the Hamiltonian side.
4. *Rotations:* for a fixed α, $Q = q\cos\alpha + p\sin\alpha$ and $P = -q\sin\alpha + p\cos\alpha$ — the time-α map (6.2) of the oscillator $\frac{1}{2}(q^2 + p^2)$ — is canonical, and

$$F_1 = \frac{2qQ - (q^2 + Q^2)\cos\alpha}{2\sin\alpha}, \qquad F_2 = \frac{2qP + (q^2 + P^2)\sin\alpha}{2\cos\alpha} \tag{8.18}$$

   generate it where $\sin\alpha \neq 0$ and where $\cos\alpha \neq 0$ respectively. For F_1, the relations (8.16) give

$$p = \frac{\partial F_1}{\partial q} = \frac{Q - q\cos\alpha}{\sin\alpha}, \qquad P = -\frac{\partial F_1}{\partial Q} = \frac{Q\cos\alpha - q}{\sin\alpha}.$$

   Solving the first relation gives $Q = q\cos\alpha + p\sin\alpha$, and substituting it in the second gives $P = -q\sin\alpha + p\cos\alpha$. For F_2 the same computation uses (8.17), in the independent variables (q, P).

Where each type exists

The rotation illustrates how the domains of the two types differ. Since $\partial Q/\partial p = \sin\alpha$ and $\partial P/\partial p = \cos\alpha$, the type-1 coordinates (q, Q) fail exactly on $\alpha \in \pi\mathbb{Z}$, including the identity at $\alpha = 0$. The type-2 coordinates (q, P) fail exactly on $\alpha \in \frac{\pi}{2} + \pi\mathbb{Z}$, including the swap at $\alpha = \pi/2$. At each of those angles the *other* formula in (8.18) is available, and it reduces to the generating function already given: $F_2 = q \cdot P$ at $\alpha = 0$, and $F_1 = q \cdot Q$ at $\alpha = \pi/2$. Figure 8.2 draws the collapse.

8.5 Time-dependent canonical maps

A generating function may depend on t; then the Hamiltonian changes, and the phase-space action says exactly how.

Theorem 8.12 (The new Hamiltonian). *Let $F_2(q, P, t)$ be C^∞ with $\det(\partial^2 F_2/\partial q\, \partial P) \neq 0$. By Proposition 8.10 the relations (8.17) determine a canonical map near each point at each fixed t;* fix a branch, *that is an open $D \subseteq \mathcal{M} \times \mathbb{R}$ on which (8.17) defines a single-valued C^∞ map $(q, p, t) \mapsto (Q, P)$, a diffeomorphism onto its image at each t. (Non-degeneracy*

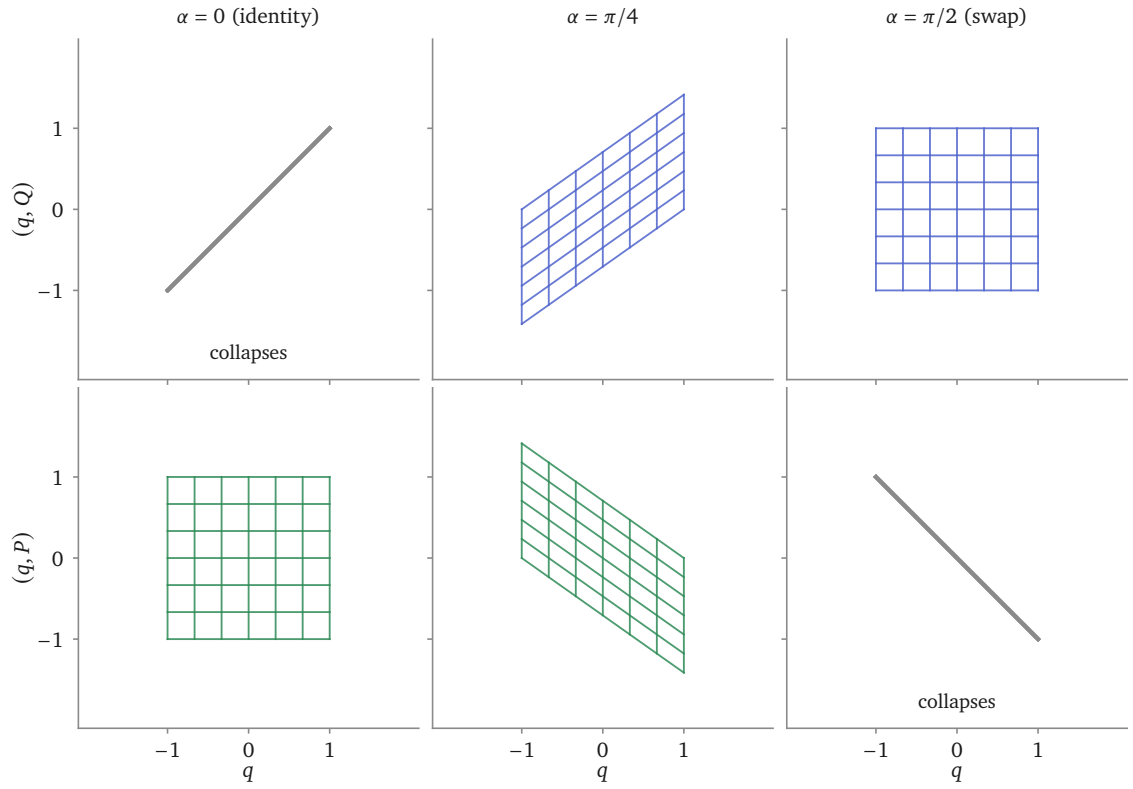


Figure 8.2: A square grid of the (q, p) plane under $(q, p) \mapsto (q, Q)$ (top) and $(q, p) \mapsto (q, P)$ (bottom), for the rotation by α. Where $\sin\alpha$ or $\cos\alpha$ vanishes the grid collapses to a line, and that type of generating function does not exist.

alone gives only local inverses.) Let $H(q, p, t)$ be a Hamiltonian and put

$$K = H + \frac{\partial F_2}{\partial t}, \tag{8.19}$$

the partial derivative taken at fixed (q, P) and the right side expressed in (Q, P, t) on the image of D. Then a C^1 path $t \mapsto (q(t), p(t))$ that stays in D satisfies Hamilton's equations for H if and only if its image $t \mapsto (Q(t), P(t))$ satisfies Hamilton's equations for K.

Proof. Along any C^1 path, differentiating $F_2(q(t), P(t), t)$ and using (8.17),

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(F_2 - Q\cdot P\big) = p\cdot\dot q + Q\cdot\dot P + \frac{\partial F_2}{\partial t} - \dot Q\cdot P - Q\cdot\dot P = p\cdot\dot q - P\cdot\dot Q + \frac{\partial F_2}{\partial t},$$

so with (8.19)

$$\big(p\cdot\dot q - H\big) - \big(P\cdot\dot Q - K\big) = \frac{\mathrm{d}}{\mathrm{d}t}\big(F_2 - Q\cdot P\big). \tag{8.20}$$

Integrate over $[t_0, t_1]$: the two phase-space actions differ by a boundary term $[\,F_2 - Q\cdot P\,]_{t_0}^{t_1}$, a function of the endpoint states alone. A variation of the path vanishing at both ends leaves that boundary term unchanged, so the two actions have the same first variation under every such variation; and because the map is a diffeomorphism at each t on D, a C^1 variation of (q, p) inside D vanishing at the ends corresponds to a C^1 variation of (Q, P) vanishing at the ends, and conversely.

It remains to say that vanishing-end stationarity is Hamilton's equations. By (4.11) the first variation of $\int (p\cdot\dot q - H)$ in the directions (η, ζ) is $\int [\zeta\cdot(\dot q - H_p) - \eta\cdot(\dot p + H_q)]\mathrm{d}t$ when η

vanishes at the ends; taking $\eta = 0$ and then $\zeta = 0$, Lemma 3.3 makes it vanish for all C^2 variations vanishing at the ends exactly when $\dot q = H_p$ and $\dot p = -H_q$. The same statement on the (Q, P) side, with K, finishes the proof. ∎

Why is K not simply H written in the new variables? The map itself changes with t. Even with $H = 0$, a state that does not move has moving new coordinates. The term $\partial F_2/\partial t$ is the Hamiltonian of that motion, as the next example shows, where it is $-\Omega L_3$ and turns the frame. It vanishes when F_2 has no explicit time dependence, and then Proposition 8.3 applies.

Example 8.13 (The rotating frame)**.** In the plane, let $R(\Omega t)$ be the anticlockwise rotation by Ωt and take

$$F_2(q, P, t) = q \cdot R(\Omega t)P. \tag{8.21}$$

Then $p = R(\Omega t)P$ and $Q = R(\Omega t)^{\mathsf T} q = R(-\Omega t)q$: the new position is the old one read in a frame turning at the angular speed Ω, and the momentum turns with it. Differentiating at fixed (q, P) gives

$$\frac{\partial F_2}{\partial t} = \Omega q \cdot \begin{pmatrix} -p_2 \\ p_1 \end{pmatrix} = -\Omega(q_1 p_2 - q_2 p_1) = -\Omega L_3,$$

where $L_3 = q_1 p_2 - q_2 p_1$. Thus

$$K = H - \Omega L_3. \tag{8.22}$$

Take a potential that is *fixed in the rotating frame*: $H = |p|^2/2m + U\big(R(-\Omega t)q\big)$. Both $|p|$ and L_3 are unchanged by a rotation, so in the new variables

$$K = \frac{|P|^2}{2m} + U(Q) - \Omega\,(Q_1 P_2 - Q_2 P_1),$$

which does not contain t and is therefore conserved. It is not the energy.

The choice of F_2 can be used to solve the motion. If $K \equiv 0$, then $\dot Q = \dot P = 0$, so every motion has constant new coordinates. The condition on F_2 is the Hamilton–Jacobi equation, developed in Lecture 9.

Summary.

- *Handed on:* canonical maps and their tests (Theorem 8.2); closed and exact 1-forms and the Poincaré lemma (Theorem 8.7); F is canonical if and only if $p \cdot \mathrm dq - P \cdot \mathrm dQ$ is closed (Theorem 8.8); the generating functions F_1 and F_2 (Theorem 8.9 and Proposition 8.10); and $K = H + \partial_t F_2$ (Theorem 8.12).
- *Next:* which F_2 makes $K \equiv 0$? Lecture 9 answers: a solution of the Hamilton–Jacobi equation.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, §6.5 (pp. 174–176): (6.5.3)–(6.5.4) are the closedness criterion, (6.5.7) is (8.16), (6.5.8) is the point-transformation example in Example 8.11, and Exercise 6.5-1 is the singularity of the identity. Arnold, *Mathematical Methods of Classical Mechanics*, §48 (pp. 266–268): generating functions, with our signs; his ω is $\mathrm dp \wedge \mathrm dq$, which nothing here uses. Tong, *Lectures on Classical Dynamics*, §4.4 (pp. 100–104).